

SPARSH ROY

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EDUCATION

Hopewell Valley Central High School

Princeton, NJ

High School Diploma · GPA: 4.25 (Weighted) · Expected Spring 2028

- Activities: 25–26 President; FRC Robotics (Spirit Captain, Programmer, Strategy); Investment Club Founder & VP; Hope Squad Founding Member & Secretary; Competitive Programming Team Founder; Science Olympiad; Mu Alpha Theta; Youth Environmental Society; PreMed Club; JV Policy Debate; Model UN & Congress; Boys Volleyball; Ultimate Frisbee

MIT Beaver Works Summer Institute

Cambridge, MA

Program: Cognitive Assistants

- COMPLETED prestigious computer science program working with MIT professors on AI systems and cognitive assistant technologies

Mercer County Community College

Trenton, NJ

Dual Enrollment

- TAKING advanced coursework in Calculus III, Linear Algebra, and Business Law

WORK EXPERIENCE

MIT Computer Science & Artificial Intelligence Laboratory (CSAIL)

Cambridge, MA

Research & Development – Kellis Lab

Present

- CONDUCTING research and development within the Kellis Lab at MIT CSAIL, contributing to cutting-edge computational biology and AI-driven genomics projects

Velvt

Princeton, NJ

Chief Operating Officer & Co-Founder

April 2026 – Present

- ARCHITECTED end-to-end operational infrastructure for a local-first behavioral intelligence system that passively models focus patterns and attention fragmentation entirely on-device; no cloud dependency
- SCALED company from private beta to production, overseeing cross-functional execution across product, engineering, and go-to-market; built growth systems and privacy architecture coordination
- DROVE operational strategy ensuring Velvt's core principles—neutrality, local-first privacy, and low-noise behavioral insight—are preserved at every layer of the product

DreamIT

Berkeley, CA

Lead Founding Engineer & Co-Founder

November 2025 – Present

- BUILT AI-driven software platform owning end-to-end frontend and backend architecture; optimized systems to deliver seamless, high-performance user experience across web and mobile
- COLLABORATED with founders on product strategy and roadmap decisions, translating product vision into scalable technical solutions with clean architecture
- PARTICIPATED in venture capital meetings and pitch sessions, gaining direct exposure to fundraising, financial modeling, and investor decision-making processes

SPIKE Robotics – FRC Team 293

New Jersey

Chief Sustainability Officer / Strategy Co-Lead / Software Engineer

August 2024 – Present

- DEVELOPED robot control systems, autonomous routines, and technical tooling in a fast-paced engineering environment under strict competition deadlines; progressed from SWE to CSO over 1 yr 10 mo
- LED sustainability and long-term resource planning, sponsorship organization, and documentation systems to support scalable team infrastructure across competition seasons

AidSense

Jersey City, NJ

Chief Financial Officer

May 2025 – Present

- LEAD financial operations, modeling, and strategic planning for the Congressional App Challenge-winning crisis resource platform following its recognition by the US House of Representatives

Institute for Youth in Policy

Remote

Technology Manager (promoted from Innovative Projects & Technology Intern)

July 2025 – May 2026

- LED technical execution of research and policy infrastructure supporting capstone projects and publications across multiple policy verticals
- DEVELOPED internal tech automations to streamline research intake, review cycles, and publication pipelines; reduced turnaround time by 40%
- COORDINATED with research leads, editors, and executive leadership to translate program needs into technical solutions

AWARDS & PROJECTS

NJ INNOVATE100 Top Innovator

May 2026

- NAMED to the 2026 INNOVATE100 list by Innovate100 (sponsored by Rutgers, Princeton & Nokia), recognizing catalysts of New Jersey's innovation ecosystem; one of the youngest recipients selected for sustained leadership in software engineering and entrepreneurship

2025 Congressional App Challenge Winner – US House of Representatives

December 2025

- DEVELOPED AidSense, a Flutter app using AI to help individuals in crisis find local resources; features include offline access, Firebase authentication, ChatGPT-powered assistance, and interactive maps. Showcased at US Capitol #HouseOfCode 2026

2026 Hackathon Finalist – 1435 Capital (\$2.3B AUM)

January 2026

- DESIGNED FLUX, an AI-driven housing supply platform addressing New Jersey's \$50B housing shortage; placed as finalist among 200+ teams. Led technical architecture, AI/ML systems design, and economic modeling; projected 56,000 units over three years with self-sustaining financial model

Fundraising Campaign Organizer – Leukemia and Lymphoma Society

- RAISED \$10,000 to support blood cancer research, patient services, and advocacy as part of Team Thrive

SKILLS & CERTIFICATIONS

Technical Skills: Python, JavaScript, TypeScript, React, Flutter/Dart, Node.js, AI/ML, Data Analysis, Firebase, AWS, Financial Modeling, Excel, PowerPoint, Adobe Suite

Certifications: CYIA 98% Scorer (Young Investors Society) · Strategic Negotiations (UC Davis) · Corporate Finance (Columbia) · Probability (Harvard) · Investment Banking (JPMorgan Chase) · Cloud Security (AWS) · Cybersecurity (Mastercard)